

Republic of the Philippines

# Laguna State Polytechnic University

Province of Laguna

ITEP 414 - System Administration and Maintenance

## Enterprise Server Deployment - Installation Guide

|                |                                                      |
|----------------|------------------------------------------------------|
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| Section        | BSIT-4E                                              |
| Course / Week  | ITEP 414 - Week 3                                    |
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Portfolio Project: Junior System Administrator Deployment Documentation

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# 1. Project Overview

ABC Startup Solutions plans to deploy its first Linux server for future file sharing, remote administration, database hosting, web hosting, and internal services. As the Junior System Administrator, the task is to install Ubuntu Server, configure the required baseline settings, verify that the server works, and document the process so another administrator can reproduce the deployment.

This guide follows the supplied Week 3 module. Where the module requires values that can only come from the actual virtual machine - such as the DHCP-assigned IP address, exact partition scheme, filesystem, and screenshots - this document identifies the required evidence instead of inventing it.

## 2. Objectives

- Install Ubuntu Server in a virtual machine using the required enterprise baseline.
- Configure language, keyboard, networking, hostname, user account, disk, and OpenSSH.
- Verify login, hostname, IP address, Internet connectivity, updates, and SSH service status.
- Compare BIOS and UEFI and explain the Ubuntu boot sequence.
- Install Windows Server Evaluation for the bring-home activity and compare Windows Server, Ubuntu Server, and Rocky Linux.
- Produce professional documentation suitable for a GitHub portfolio.

## 3. Software Requirements

- Oracle VirtualBox or VMware Workstation.
- Ubuntu Server LTS ISO - latest stable version.
- Windows Server Evaluation ISO for the bring-home activity.

## 4. Virtual Machine Specifications

| Component           | Required value                                                                    |
|---------------------|-----------------------------------------------------------------------------------|
| Name                | Ubuntu-Server-Week03                                                              |
| RAM                 | 4 GB                                                                              |
| CPU                 | 2 virtual processors                                                              |
| Storage             | 40 GB virtual disk (VDI/VMDK)                                                     |
| Network             | NAT, or Bridged if instructed                                                     |
| Optical Drive       | Ubuntu Server ISO                                                                 |
| Hostname            | server01                                                                          |
| Administrative user | gayoso - based on the module example "yourlastname"; verify against the actual VM |
| IP address          | DHCP unless instructed otherwise; record actual assigned address                  |

## 5. Step-by-Step Ubuntu Server Installation Procedure

### 5.1 Create the virtual machine

Create a new VM named **Ubuntu-Server-Week03**. Assign 4 GB RAM, 2 virtual processors, a 40 GB VDI/VMDK disk, NAT networking (or Bridged if instructed), and attach the Ubuntu Server ISO.

#### ACTUAL VM SCREENSHOT REQUIRED

Create the virtual machine evidence  
Suggested file: screenshots/01-vm-settings.png

### 5.2 Language

Start the installer and select **English**.

#### ACTUAL VM SCREENSHOT REQUIRED

Language evidence  
Suggested file: screenshots/02-language-keyboard.png

### 5.3 Keyboard Layout

Select the appropriate keyboard layout, such as English (US).

#### ACTUAL VM SCREENSHOT REQUIRED

Keyboard Layout evidence  
Suggested file: screenshots/02-language-keyboard.png

### 5.4 Network Configuration

Accept DHCP unless instructed otherwise. Record the assigned IP address.

#### ACTUAL VM SCREENSHOT REQUIRED

Network Configuration evidence  
Suggested file: screenshots/03-network-installer.png

### 5.5 Hostname and User Account

Set the hostname to **server01**. Create a non-root administrative user. This portfolio uses **gayoso** based on the module username example; the actual VM value should be verified. Use a strong password.

#### ACTUAL VM SCREENSHOT REQUIRED

Hostname and User Account evidence  
Suggested file: screenshots/05-profile-hostname.png

### 5.6 Disk Partition

Choose **Guided - Use Entire Disk**. Record the actual partition scheme, filesystem used, and 40 GB disk size shown by the installer.

#### ACTUAL VM SCREENSHOT REQUIRED

Disk Partition evidence  
Suggested file: screenshots/04-storage-layout.png

## 5.7 SSH Server

Enable **Install OpenSSH Server**.

**ACTUAL VM SCREENSHOT REQUIRED**  
SSH Server evidence  
Suggested file: screenshots/06-openssh-selection.png

## 5.8 Additional Packages

Do not install additional packages unless instructed.

**ACTUAL VM SCREENSHOT REQUIRED**  
Additional Packages evidence  
Suggested file: screenshots/06-openssh-selection.png

## 5.9 Complete Installation

Finish the installation, restart the virtual machine, and remove the installation ISO if prompted.

**ACTUAL VM SCREENSHOT REQUIRED**  
Complete Installation evidence  
Suggested file: screenshots/07-install-complete.png

## 6. Configuration Details

| Setting             | Required / recorded value                      |
|---------------------|------------------------------------------------|
| Language            | English                                        |
| Keyboard            | Appropriate keyboard layout, e.g. English (US) |
| Network             | DHCP unless instructed otherwise               |
| Hostname            | server01                                       |
| Full name           | Cedrick Gayoso                                 |
| Username            | gayoso - verify actual VM                      |
| Disk                | 40 GB - Guided - Use Entire Disk               |
| Partition scheme    | Record actual installer result                 |
| Filesystem          | Record actual filesystem used                  |
| SSH                 | OpenSSH Server installed during setup          |
| Additional packages | None unless instructed                         |

## 7. Server Verification

After the first successful boot, perform all six verification tasks required by the module. Capture actual screenshots from the VM.

### 7.1 Login

Log in using the account created during installation.

**ACTUAL VM SCREENSHOT REQUIRED**  
Successful login  
Suggested file: screenshots/08-login.png

### 7.2 Verify Hostname

Run: `hostname`

Required hostname: **server01**.

**ACTUAL VM SCREENSHOT REQUIRED**  
hostname showing server01  
Suggested file: screenshots/09-hostname.png

### 7.3 Verify IP Address

Run: `ip addr`

Record the actual DHCP-assigned IPv4 address. The project PDF does not provide a fixed IP address.

**ACTUAL VM SCREENSHOT REQUIRED**  
ip addr showing actual IP  
Suggested file: screenshots/10-ip-address.png

### 7.4 Test Internet Connectivity

Run: `ping -c 4 google.com`

Capture the successful response.

**ACTUAL VM SCREENSHOT REQUIRED**  
successful ping response  
Suggested file: screenshots/11-ping-google.png

### 7.5 Update the Server

Run:

`sudo apt update`

`sudo apt upgrade -y`

**ACTUAL VM SCREENSHOT REQUIRED**  
update and upgrade process  
Suggested file: screenshots/12-apt-update-upgrade.png

### 7.6 Verify SSH Service

Run: `systemctl status ssh`

The service should display **active (running)**.

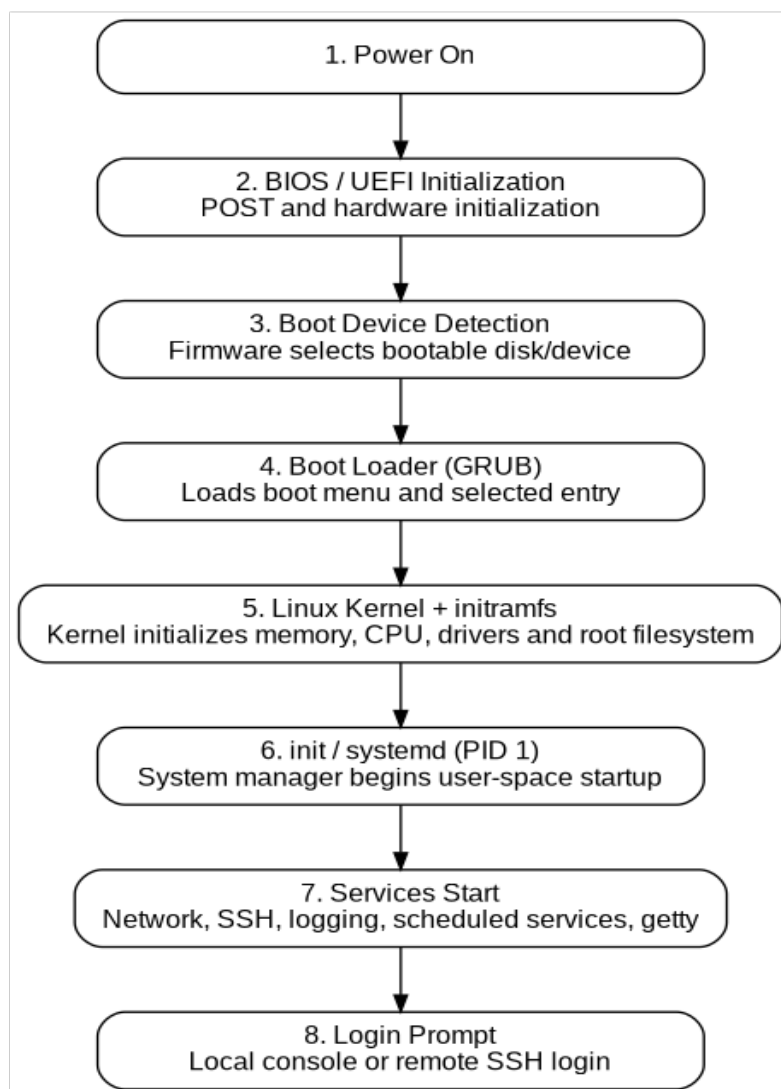
**ACTUAL VM SCREENSHOT REQUIRED**  
SSH active (running)  
Suggested file: screenshots/13-ssh-status.png

## 8. BIOS vs UEFI Comparison

The module requires a comparison covering definition, boot process, maximum disk support, partition style, security features, speed, advantages, disadvantages, and modern usage. See **BIOS\_vs\_UEFI.pdf** for the complete table and explanation.

## 9. Ubuntu Boot Process Flowchart

Required sequence: Power On -> BIOS/UEFI Initialization -> Boot Device Detection -> Boot Loader (GRUB) -> Linux Kernel -> init/systemd -> Services Start -> Login Prompt.



## 10. Windows Server Installation Summary

The bring-home activity requires installing Windows Server Evaluation in a new virtual machine. Minimum tasks are: install the operating system, assign a computer name, create an Administrator password, log in successfully, and capture installation screenshots.

| Item                   | Recorded value / status                                                           |
|------------------------|-----------------------------------------------------------------------------------|
| Operating system       | Windows Server Evaluation                                                         |
| Computer name          | Not specified by the module - record the actual name assigned during installation |
| Administrator password | Created during installation; do not publish the password                          |
| Login                  | Must be verified with an actual screenshot                                        |
| Screenshots            | Required from the actual Windows Server installation                              |

**ACTUAL VM SCREENSHOT REQUIRED**  
Windows Server installation / login evidence  
Suggested file: screenshots/14-windows-server-installation.png

## 11. Operating System Comparison

The module requires comparison of Windows Server, Ubuntu Server, and Rocky Linux using licensing, user interface, package management, security, performance, typical enterprise use cases, advantages, and disadvantages. See **OS\_Comparison.pdf** for the full table and discussion.

## 12. Problems Encountered and Solutions Implemented

Record only problems that actually occurred in the laboratory. No VM logs or screenshots were supplied with the assignment PDF, so this package does not fabricate problems or solutions. Use the format below after completing the lab.

| Problem encountered | Solution implemented                     |
|---------------------|------------------------------------------|
| Actual issue 1      | Actual troubleshooting action and result |
| Actual issue 2      | Actual troubleshooting action and result |

## 13. Lessons Learned

The deployment process shows that a server installation should be treated as a controlled and documented procedure. The required baseline settings create a repeatable starting point, while the verification commands provide evidence that identity, networking, updates, and remote administration are working. The BIOS/UEFI and boot-process activities also provide a structured way to understand where startup failures can occur. Finally, the comparison of Windows Server, Ubuntu Server, and Rocky Linux demonstrates that operating-system selection should be based on workload, administration, security, performance, and enterprise requirements.

## 14. References

- Laguna State Polytechnic University, ITEP 414 - System Administration and Maintenance, Week 3 Portfolio Project: Enterprise Server Deployment and Operating System Installation.
- Official Ubuntu Server documentation for implementation details.
- Official Microsoft Windows Server documentation for the bring-home activity.
- Official Rocky Linux documentation for operating-system comparison research.
- UEFI Forum documentation/specifications for firmware research.